HISTORICAL DEVELOPMENT OF SCIENCE & TECHNOLOGY in MESOAMERICA

OLMEC | MAYAN | AZTEC | INCA CIVILIZATION

MESOAMERICA

It is a Greek word which means “Middle/Central America”.

A region and cultural area in the Americans, extending from Central Mexico to Belize, Guatemala, El Salvador, Honduras, Nicaragua and northern Costa Rica.

OLMEC CIVILIZATION

The name Olmec is a **Nahuatl** – the Aztec language - word meaning the rubber people; the Olmec made and traded rubber throughout Mesoamerica.First people to figure out how to convert latex of the rubber tree into something that could be shaped, curved and hardened.



Ang katagang **OLMEC** ay nangangahulugang **RUBBER PEOPLE** dahil sila ang kauna-unahang gumamit dagta ng mga punong rubber o goma.

Great pyramid in La Venta, Tabasco





The Olmec heartland, where the Olmec reigned

San Lorenzo is located in Veracruz State, about 38 miles (60km) from the Gulf of Mexico.





Tres Zapotes is a Mesoamerican archaeological site located in the south-central Gulf Lowlands of Mexico in the Papaloapan River plain. Tres Zapotes is sometimes referred to as the third major Olmec capital (after San Lorenzo Tenochtitlán and La Venta).

Where did they live? The Olmec lived along the Gulf Coast of modern-day Mexico in tropical rain forests and lowlands from around 1200 BCE to 400 BCE.

Who did they influence? Both the Maya and the Aztecs were influenced by the Olmec civilization, the earliest known civilization in Mesoamerica

Who was in charge? Sculptures and temples indicate that kings or priests led the Olmec society.

What did the religious centers look like?

At former religious centers characterized by pyramid-shaped temples, giant stone heads weighing 10 tons stood up to 11 feet tall.

Olmec Contributions? The Olmec also developed a form of writing, as well as a calendar

What allowed the Olmec to spread their culture? Through their trade networks, the Olmec were able to spread many characteristics of their culture, religion, architecture, and social structure spread north to the Valley of Mexico and around Central America.

OLMEC CIVILIZATION

1. RELIGION : There are no direct written accounts of Olmec beliefs, but their notable artworks provide clues about their religion. Some deities are as follows:

- The Maize deity -

- The Rain Spirit / Were Jaguar -
- The fish or Shark Monster -
- The Bird Monster -
- The Olmec Dragon -

2. <u>TECHNOLOGY AND ART</u>: They created a hydraulic system (a machine that is operated by the force offered or the pressure given out when liquid is forced through a small opening of a tube), and canals that served as giant waterways for trade. Even some of their art forms influenced other early Mesoamerican people. **One of the Olmec signature art mediums was blue jade.**

3. <u>PRACTICES</u> : First to play with **balls** made from **solid rubber** and that they may have **practiced ritual bloodletting.**

4. <u>COLOSSAL HEADS</u> : All portray mature men with fleshy cheeks, flat noses, and slightly crossed eyes. However, **none of the heads are alike,** and each boasts a unique headdress, which suggests **they represent specific individual**

5. <u>WRITING SYSTEM</u> : Relatively little was known about Olmec language and writing before the discovery of the **Cascajal block**. The Olmec certainly had symbols: **"It's the first group of people to have elaborate symbolism,"** explained Houston. In 2002, a roller stamp and fragments of a plaque were found in San Andres, Tabasco, which dated to at least 650 BC. **They also had a calendar system**, but there is limited evidence to describe the system in detail.



The block measures to over a foot in length and is about the size of a small flat screen TV, but much thicker. It features **62 symbols arranged in linear patterns on one side of the serpentine slab.**

5. <u>ENGINEERING</u> : The Olmec were the **first great engineers of Mesoamerica.** There is an aqueduct at San Lorenzo, carved out of dozens of massive stones then laid side-by side. The royal compound at La VENTA shows engineering as well: the "massive offerings" of **Complex A** are complicated pits filled with stones, clay, and supporting walls, and there is a tomb there built with basalt support columns.

6. <u>TRADE AND COMMERCE</u> : The Olmecs created long-distance trade routes to obtain the things they needed, eventually making contacts all the way from the valley of Mexico to Central America. Olmec traders swapped finely made **Olmec celts, masks and other small pieces of art** with other cultures such as the Mokaya and Tlatilco, getting jadeite, serpentine, obsidian, salt, cacao, pretty feathers and more in return.

<u>DECLINE OF THE OLMEC CIVILIZATION</u>

Olmec systematically destroyed their ceremonial centers at both San Lorenzo and La Venta and then deserted the sites. Some speculate reasons involving civil conflicts or doubts about the effectiveness of the ruling classes.

<u>MAYAN CIVILIZATION</u>

One of the famous civilizations that lasted for approximately 2000 years. The Maya made important discoveries in the areas of science and cosmology which enabled them to create a complex calendar system.

Where are they located? The Maya lived in modern-day southern Mexico and Central America. including the areas that are today Belize, El Salvador, Guatemala, and Honduras.

What do they grow? They were an agriculture-based society. They grow corn, beans, and squash, and practice many of the same crafts, such as weaving and pottery.

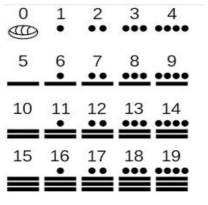
How did their location help them? Their central location made it very easy for the Maya to trade and interact with other cultures from North and South America.

How did they trade? The Maya built a large and complex system of roads to stay connected with other cities and peoples.

What is left behind? Mayan ruins include huge ziggurat and observatories used by astronomers.

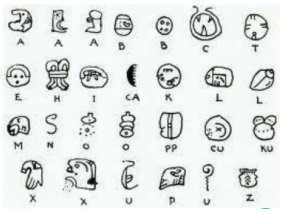
1. ASTRONOMY: There were just **365 days** in the **Haab year**, they knew that a year was slightly longer than 365 days, calculating it to be 365.2420 days (**the true approximation is 365.2422**).

- Workout that **81 lunar months constituted 2,392 days.**
- Length of the lunar month at **29.5308 days**, astoundingly close to the modern estimation of 29.53059 days.
- Worked out the **584-day cycle of Venus** with a difference of two hours.



2. MATHEMATICS : The Maya had an extremely accurate numerical system. It had only three symbols: **0 (shell shape), 1 (a dot) and 5 (a bar).** They used these three symbols to express numbers from 0 to 19; numbers after 19 were composed vertically in groups of 20, using place markers.

3. WRITING SYSTEM : The Maya invented the most advanced form of writing, known as “glyphs.” History suggests that the Maya used around 700 different glyphs, and astonishingly 80 percent of the language is still understood today.



4. **MAYAN CALENDAR:** uses three distinctive dating frameworks.

- The **Tzolkin** joins a cycle of 20 named days with another cycle of 13 numbers to deliver 260 distinct days.
- The **Haab** was the sun-based Calendar with 365 days. It was made up of a year and a half with 20 days in each month, and five additional days which were known as **Wayeb and were considered to represent a dangerous time.**
- The **Long Count** was a non-repeating calendar starting from the beginning of the Mayan period.

Mayan Long Conversions

Bakun = 144,000 days
Katun = 7,200 days
Tun = 360 days
Uinal = 20 days
Kin = 1 day

5. RUBBER : They realized it was strong and versatile and began to make and use water-resistant cloth, glue, bindings for books, figurines, and rubber balls.

6. LAW & ORDER : In the Mayan Empire, laws were standardized across every state and were applicable to all levels of society. Pok-A-Tok , a Mayan ballgame, is used as a way to settle conflicts between warring groups and noblemen.

7.MAYAN ART : The Mayan built looms of weaving cloth and devised a rainbow of

glittery paints made from a mineral called **mica**.

8.TECHNOLOGY: They built hydraulics systems with sophisticated waterways to supply water to different communities.

DECLINE OF MAYAN CIVILIZATION

Remains their greatest mystery. Their civilization was not destroyed by an overwhelming outside force but disintegrated from within. Many hypotheses have been proposed: overpopulation, famine, epidemics, and civil disorder.

AZTEC CIVILIZATION

The Aztecs had a reputation for making human sacrifices in elaborate rituals to appease their many gods. Eventually, they settled in the Valley of Mexico and founded their capital, **Tenochtitlan**, in 1345 and at the beginning of the 16th century it was one of the largest cities in the world.

Where did they live? Aztec people settled in the Valley of Mexico and what is now Mexico City.

Where was the Capital?The capital city of Tenochtitlan approximately 30 million living in the entire empire.

What was another name for the Aztec people? Toltec Empire

What were the Aztecs known for? The Aztecs were fierce warriors who used military power to build a huge empire.

How did they become rich? Collecting taxes from the people they conquered was how the Aztec became so wealthy.

1. MANDATORY EDUCATION: Children are mandated to get education regardless of their social class, gender or age.

- Boys had to undergo the additional hardship of exposure to extreme temperatures, designed to help them develop yolteotl, a heart of stone, which was part of their warrior training.

2. CHOCOLATES : The Aztec in Mexico developed chocolate during their time. In the Mayan culture, they used it as currency. The Aztec valued the cacao beans highly and made it as part of their tribute to their gods.

3. ANTISPASMODIC MEDICATION: They used a type of antispasmodic medication that could prevent muscle spasm and relax muscles, which could help during surgery

4. CHINAMPAS: It is a form of Aztec technology for agricultural farming in which the land was divided into rectangular areas and surrounded by canals.

5. AZTEC CALENDAR: This enabled them to plan their activities, rituals and planting.

TWO CYCLES:

A. Tonalpohualli (counting of the days)

- It is a sacred calendar that forms a 260-day cycle
- Intermeshing one of the 20 day signs (represented by glyphs) and a number from 1 to 13 (represented by dots)

B. Xiuhpohualli (counting of the years)

- An agricultural calendar that consists of a 365-day cycle.
- Signifies when particular religious ceremonies and festivals should be held.
- 360 named days and 5 nameless days (nemontemi)

6. INVENTION OF CANOES: A light narrow boat used for travelling in water systems.

- Best way to get to and away from the city was via canoe.
- There were no pack animals or carts with wheels.

- Canoes played a crucial role to the huge success of Tenochtitlan and Tlatelolco.

DECLINE OF THE AZTEC CIVILIZATION

The arrival of Spanish conquistadors such as Hernando Cortes brought an end to the Aztecs. War and European diseases devastated the Aztec population.

INCA CIVILIZATION

Largest empire in pre-Columbian America and possibly the largest empire in the world in the early 16th century.Known for building large stone cities, beautiful temples, an advanced government, a detailed tax system, and an intricate road system.

Where did they live? The Incas developed in the Andes Mountains in what is now Peru.

Who was in charge? The Incas were ruled by an emperor who had absolute power.

What was farming like? To farm the steep land: they cut terraces into the mountainside and built aqueducts to irrigate crops.

What did they grow? The Inca grew mostly corn and potatoes.

How did the Inca move? On which runners carried messages to far cities and the capital of Cuzco.Many of these roads are still used today.

1. ROADS & COMMUNICATION: Incans did invent a network of roads and highways that connected their territory on a scale never seen before in South America. The roads were usually paved with stone. They also built bridges where the roads needed to cross rivers.

- Fast young men called "chasqui" would run from one relay station to the next.

2. QUIPU: A series of strings with knots. The number of knots, the size of the knots, and the distance between knots conveyed meaning to the Inca, sort of like writing. Only specially trained officials knew how to use quipus.

3. STONE BUILDINGS: The Inca were able to create sturdy stone buildings

4. TERRACE FARMING: They used irrigation and water storage techniques to grow crops in all sorts of terrain from the deserts to the high mountains.

5. INCA CALENDAR: Calendar with 12 months to mark their religious festivals and prepare them for planting season.

6. SUSPENSION BRIDGE: The bridges were constructed using ichu grass woven into large bundles which were very strong.

7. INCA TEXTILES: since cloth was one of the specially prized artistic achievements.

DECLINE OF THE INCA CIVILIZATION

Factors of decline include Spanish arrival in Peru, Internal conflicts and disease.

DEVELOPMENT OF SCIENCE AND TECHNOLOGY IN AFRICA (Egypt)

SCIENCE AND MEDICINE

Ancient Egyptian Doctors

- Priests were the first people to practice medicine.
- The first known physician in Egypt was the high priest and Vizier, Imhotep. They are considered by many to be the true father of medicine.

Ancient Egypt Medical Papyrus

1. Edwin Smith Papyrus -It is a medical text on surgical trauma, dating back to 1600 B.C. The papyrus documents 48 cases including injuries to the head, neck, arms and torso, along with the treatments used.

2. Ebers Papyrus - Purchased by Georg Ebers in 1873 at Luxor, the Ebers Papyrus dates back to 1500 B.C. It is a papyrus

scroll that contains over 700 magical spells and remedies.

3. Kahun Gynecological Papyrus - This text contains 34 sections that deal with specific gynecological problems, diagnosis, and treatment. This text informs that the ancient Egyptians believed that many of the maladies they suffered from occurred as a result of different conditions of the womb.

Eyes and womb are, for some reason, closely linked in ancient Egyptian health and medicine.

MATHEMATICS

Ancient Egyptian Measurements

Ancient Egyptians didn't measure things using centimetres and metres. They used **cubits**, spans and fingers. A cubit is the measurement from the tip of your longest finger to the bottom of your elbow.

A **palm** is the distance across your palm.
A **finger** is the width of your finger.

Merkhet - a timekeeping and surveying instrument, made in Ancient Egypt, 600 BCE. Made in bronze with hieroglyphic text inlaid with electrum metal. As a surveying instrument, it would have been used to layout axes of buildings.

MATHEMATICAL PAPYRUS

1. Rhind Mathematical Papyrus- is one of the best known papyrus purchased and named after Alexander Henry Rhind in 1858 in Luxor, Egypt. It consists of a long table of fractional parts to help with division, followed by the solutions of 84 specific problems in arithmetic and geometry.

2. Moscow Mathematical Papyrus - is an ancient Egyptian mathematical papyrus, also called the Golenishchev Mathematical Papyrus, after its first owner outside of Egypt, Vladimir Golenishchev. The papyrus is well known for its geometry problems.

ASTRONOMY

Calendar

The ancient Egyptians used a lunar calendar for their religious festivals and rituals.

They also used a solar calendar for their daily lives which contained 365 days per year.

Each year consisted of three, four-month seasons, which were named after significant events related to their agrarian lifestyle.

1. Akhet : The time of the floods
2. Peret: The time of growth
3. Shemu: The time of harvest

OBELISKS

Obelisks were used at entrances to temples of the sun god and were covered in an alloy called electrum, which is composed of four parts gold to one part silver.

Obelisks were symbolic of the pharaohs' divinity as well as objects of worship of the Egyptian sun god Ra. The hieroglyphic inscriptions recorded the deeds and accomplishments of the pharaoh.

SUNDIALS

Sundial, the earliest type of timekeeping device, which indicates the time of day by the position of the shadow of some object exposed to the sun's rays. As the day progresses, the sun moves across the sky, causing the shadow of the object to move and indicating the passage of time.

PYRAMIDS

THE GREAT PYRAMIDS OF GIZA

1. Great Pyramid- the oldest and largest of the three pyramids at Giza. It was built by Pharaoh Khufu who reigned from 2589-2566 B.C.

2. The Pyramid of Khafre- second tallest pyramid at Giza and contains Pharaoh Khafre's tomb.
3. **The Pyramid of Menkaure-** the southernmost pyramid at Giza.

WRITING SYSTEM

Hieroglyphics

The earliest form of writing was in the form of **hieroglyphics**, which simply put, were drawings that portrayed a story. The Egyptians used them to keep accurate records and maintain control of their empire. Scholars have discovered that there are three different classes of Egyptian hieroglyphics.

1. Phonograms are signs that represent a specific sound. Some correspond with the sounds of two or more letters.
2. Ideograms represent ideas instead of sounds.
3. Determinatives are hieroglyphs that were not spoken or translated.

COSMETICS

Make-up

Make-up originated with the ancient Egyptians, where men and women alike are used to wearing it. Some of the common cosmetics in ancient Egypt included:
1. **malachite**- a copper ore, which provided the green eye makeup color
2. **Kohl**- used to draw thick, distinctive black lines giving an almond shape to the eyes
3. **Red ochre**- which was used as rouge or lip color
4. **Henna**- which was widely used to stain the fingertips and toes

Mummification

Ancient Egyptians believed the body would physically come back to life, and the soul would return to it to continue living in the underworld. The key to having a smooth reunification was the ancient Egyptian mummification process that tried to preserve all the important parts, including the intestines.

Conclusion

For many years, Egypt has inspired artists, writers, scientists and intellectuals. The pharaohs left a huge scientific legacy. They were the first to introduce mummification, medicine, agriculture, fermentation, engineering and architecture. The ancient Egyptians were pioneers in astronomy: their expertise played an important role in determining the annual flooding of the Nile, and aligning the pyramids towards the pole star. They were very interested in observing the night sky and phenomena associated with the sun and moon. They performed mummification, which shows how skillful they were in chemistry.

DEVELOPMENT OF SCIENCE IN ASIA

ASIA

Is the biggest continent in the world and the home of many ancient civilizations.It is host to many cultural, economic, scientific, and political activities of all ages.
In the field of science, technology and mathematics, great civilizations have stood out: India and China**.**

INDIA

A huge peninsula surrounded by vast bodies of water and fortified by huge mountains in its northern borders. Developed various ideas and technologies useful in their everyday lives. Seventh largest country by area. Indus Valley Civilization

INDUS VALLEY CIVILIZATION

A bronze age civilization(3300-1300 BCE; Mature period 2600-1900 BCE). Developed

new techniques in handicraft and metallurgy. Noted for their urban planning, baked brick houses, elaborated drainage systems, and clusters of large non-residential buildings.

IRON AND IN METALLURGICAL WORKS

BRASS

Brass is a metallic alloy that is made of copper and zinc.

COPPER

- Used to create copper-bronze images such as Buddhist or Hindu/Mahayana Buddhist deities.
- Copper was also a component in the razors rsfor the tonsure ceremony. Copper does not rust or decay, they can survive indefinitely.
- Copper plate grants
- Sohgaura copper-plate

GOLD AND SILVER

- Used for making utensils for the royal family and nobilities.
- Royal family wore costly fabrics so it maybe assumed that gold and silver were beaten into thin fibers and embroidered or woven into fabrics or dress.

IRON

Army used “arrows tipped with iron” Indian arrow, iron heads and bamboo shafts

WOOTZ STEEL

Crucible steel was wootz. Wootz derives from the Tamil term for steel urukku.
Indian wootz steel was the first high quality steel that was produced.Damascus steel – this swords are characterized by distinctive pattern of banding and mottling reminiscent of flowing
water.

MEDICINE

AYURVEDA

- Ayurveda, The Science of Living or Knowledge of Life and Longevity
- Ayurveda traditions are a type of complementary or alternative medicine.
- Therapies and practices have been integrated in general wellness application and some cases in medical use.
- Ancient Ayurveda texts also taught surgical techniques including
- 8 Branches of Medicine

AYURVEDA

Great Trilogy of Ayurvedic Medicine:

- **Sushruta Samhita**- Is an ancient Sanskrit text on medicine and surgery such as plastic surgery and the removal of cataracts.
- **Charaka Samhita**- describes ancient theories on human body, etiology, and therapeutics for a wide range of diseases.
- Astanga Hridaya- “The Heart of eight Limbs”, contains about 7120 poetic verses about surgical instruments.

Three basic types of energy in Ayurvedic Philosophy:

• **Vata**, **Movement** -composed of space and air, governs breathing, blinking,muscle and tissue movement, pulsation of the heart, and all movements in the cytoplasm and cell membranes

• **Pitta, Metabolic System** - composed of fire and water. Governs digestion, nutrition, metabolism, and body temperature

• **Kapha, Structure**- composed of earth and water. Lubricates joints, moisturizes the skin, and maintains immunity

RHINOPLASTY

• aesthetically enhancing the nose by resolving nasal trauma

LITHOTOMY

• Is a surgical method for removal of calculi, stones formed inside certain organs, such as the kidneys, bladder and gallbladder that cannot exit naturally through the urinary system or urinary tract.

ASTRONOMY

CONFIGURATION OF THE UNIVERSE

- In Hindu cosmology, the universe is cyclically created and destroyed.
- Its cosmology divides time into four epochs namely:

1. Satya Yuga **2**. Treta Yuga
3. Dvapara Yuga **4**. Kali Yuga

THE SPHERICAL SELF SUPPORTING EARTH

Deal with the sphericity of the earth and the motion of the planets. Aitareya Brahmana-Ancient Indian philosophical text that states that the earth's rotation may be the cause of the Sun rising and setting.

ARYABHATIYA

- (476 CE – 550 CE)
- Believed in Spherical Earth
- Written by Aryabhata
- Aryabhatiya is divided into four padas or chapters

1. Gitikapada

• overview of astronomical findings.

2. Ganitapada

Mathematics
Arithmetic and Geometric progressions
Simple, quadratic simultaneous and indeterminate equations

3. Kalakriya

• "the reckoning of time"
• In it, Aryabhata divides up days, months and years according to the movement of celestial bodies.

4. Golapada

• The sphere
• Celestial relationship of earth and cosmos
• Rotation of earth on its axis
• Armillary sphere

ARMILLARY SPHERE

RIGVEDA

- Rig, praise and veda, knowledge
- Rigveda, (Sanskrit: "The Knowledge of Verses") also spelled Ṛgveda, the oldest of the sacred books of Hinduism,
- Ancient Indian collection of Vedic Sanskrit hymns
- 10 books with 1, 028 hymns in about 10, 600 verses. The hymns predominantly discuss cosmology and praise deities.
- Described a solar and luni-solar calendrical scheme.
- According to the Puranic tradition, Ved Vyasa compiled all the four Vedas, along with the Mahabharata and the Puranas.
- Is the major treatise of Indian astronomer and mathematician Bhaskara II

Parts

1. **Leelavati** – the book contains thirteen chapters, 278 verses, mainly arithmetic and measurement.
2. **Bijaganita** – divided into six parts, contains 213 verses and is devoted to algebra.
3. **Ganitadhyaya and Goladhyaya** – devoted topics on astronomy such as mean longitudes of the planet, diurnal motion, lunar and solar eclipses

MATHEMATICS MOHENJO-DARO RULER

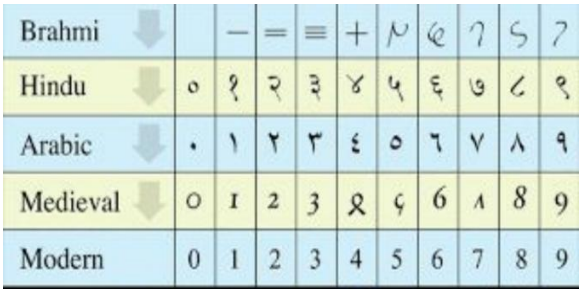
• Rulers made out of ivory
• Divided into units corresponding to 1.32 inches

ARYABHATIYA

Mathematical and astronomical treatise authored by Aryabhata.
It introduced a number of trigonometric functions, tables, techniques and algorithms of algebra

HINDU ARABIC NUMERAL SYSTEM

• Most common system for the symbolic representation of numbers in the world.
• The symbols are arranged in 3 main groups: (1) Western arabic numerals, (2) Eastern arabic numerals, and (3) Indian numerals.



Brahmi		—	=	≡	+	५	६	७	९	८
Hindu	०	१	२	३	४	५	६	७	८	९
Arabic	٠	١	٢	٣	٤	٥	٦	٧	٨	٩
Medieval	০	১	২	৩	৪	৫	৬	৭	৮	৯
Modern	0	1	2	3	4	5	6	7	8	9

<u>NUCLEAR WEAPONS AND SPACE AGENCY</u>

The roots of nuclear power in India lie in early acquisition of nuclear reactor technology from a number of western countries, particularly the American support for the Tarapur Atomic Power Station and Canada's CANDU reactors. The Indian space program received only financial support from the Soviet union, which helped the Indian Space Research Organization achieve aims such as establishing the Thumba Equatorial Rocket Launching Station, launching remote sensing satellites, developing India's first satellite---Aryabhata, and sending astronauts into space.

CHINA

- China came from the name"Qin". The first dynasty of China.
- Developed around 5000 years ago. The world's populous country with 1.404 billion.
- The third largest country by area at 9,706,961 km (3,747,879 sq. miles)

FOUR GREAT INVENTIONS OF CHINA

1. PAPER

• Inventor: Cai Lun
• Date: circa 50 – 121 A.D.
• Overview: Despite being present in China since 2nd Century B.C, Cai Lun, a court eunuch, invented the pulp papermaking process and established the use of new materials in making paper

2. PRINTING

• Inventor: Bi Sheng
• Date: circa 960 – 1279 A.D.
• Overview: The earlier form of printing, woodblock, was considered time-consuming and expensive. Bi Sheng's method of gluing carved characters into an iron plate was thought to be more efficient, rapidly spreading across Europe, leading to the Renaissance, and later all around the world

3. GUNPOWDER

- Alchemists; Zeng Gongjiang, Ding Du, and Yang Weide(credited as the writers of the earliest known recorded recipes for gunpowder).
- Date: circa 618 – 907 A.D.
- Overview: Created by Taoist alchemist in a trial to find a potion to gain human immortality by mixing elemental sulphur, charcoal, and saltpetre. Interestingly, Chinese used this discovery mainly for firecrackers while Europeans created

cannons and guns, dominating China in the mid-1800s.

4.COMPASS

- Date: 1100 AD
- The Chinese by the Han Dynasty (202 BC – 220 AD) began using north-south oriented lodestone ladle-and-bowl shaped compasses for Divination And Geomancy and not yet for navigation.
- Song dynasty writer Zhu Yu (fl. 12th century) was the first to mention use of the compass specifically for navigation at sea in his book published in 1119.

FARMING AND AGRICULTURE

DEEPWATER DRILLING

-The Chinese discovered and made extensive use of deep drilled groundwater for drinking.

-The Chinese text The Book OF Changes, originally a divination text of the Western Zhou dynasty (1046 -771 BC), contains an entry describing how the ancient Chinese maintained their wells and protected their sources of water.

RICE CULTIVATION

-It is likely that demonstrated rice was cultivated in the middle Yangtze Valley by 7000 BC, as shown in finds from THE PENGTOUSHAN culture at Bashidang, Changde, Hunan.

-By 5000 BC, rice had been domesticated at Hemudu Culture near the YangtzeDelta and was being cooked in pots.

WET FIELD CULTIVATION

-Wet field cultivation, or the paddy field, was developed in China.

-The earliest paddy field dates to 6280 BP, based on carbon dating of the grains of rice and soil ORGANIC matter found at the Chaodun site in Kushan County.

MEDICINE

ACUPUNCTURE

-The Traditional Chinese medicinal practice of inserting needles into specific points of the body for therapeutic purposes and relieving pain

-The oldest known acupuncture sticks made of gold, found in the tomb of Liu Sheng (d. 113 BC), dated to the Western Han (203 BC – 9 AD).

MOXIBUSTION

-A traditional Chinese medicine technique that involves the burning of mugwort- a small, spongy herb, used to facilitate healing.

-It can treat pains and aches and is even recognized to relieve menstrual cramps.

TUI NA MASSAGE

• The term tui na, which literally means “pinch and pull”, refers to a wide range of TCM therapeutic massage and bodywork.

• It stimulates the flow of qi to promote balance and harmony within the body using many of the same principles of acupuncture.

CHINESE HERBS

The aims of Chinese herbal therapy are to help regain balance in the body and to strengthen the body’s resistance to disease. Chinese herbs may be used to decrease cold/flu symptoms, improve your breathing, decrease pain etc.

Herbs commonly used are gingko biloba, astragalus, red yeast rice, ginger, cinnamon etc,

ARCHITECTURE

DRY DOCKS

The use of dry docks in China goes at least as far back as the 10th century A.D. In his book Dream Pool Essays, the Song dynasty polymath **Shen Kuo wrote** of dry docks for repairing ships.

PONTOON BRIDGE

- The Zhou Dynasty Chinese text of the Shi Jing (Book of Odes) records that King Wen of Zhou was the first to create a pontoon bridge in the 11th century BC.
- However, the British scientist, Sinologist, and historian Joseph Needham has pointed out that in all likely scenarios, the temporary
- The pontoon bridge was invented during the 9th or 8th century BC in China, as this part was perhaps a later addition to the book (considering how the book has been edited up until the Han Dynasty, 202 BC–220 AD).

SUSPENSION BRIDGES USING IRON CHAINS

-The first written evidence of iron chain suspension bridges comes from a local history and topography of Yunnan written in the 15th century, which describes the repair of an iron chain bridge during the reign of the Yongle Emperor (r. 1402–1424); although it is questionable if Ming Dynasty (1368–1644).

-Chinese claims that iron chain suspension bridges existed since the Han Dynasty, their existence in the 15th century predates that of anywhere else

GREAT WALL

-Chinese name: 长城 (Chángchéng /channg-chnng/ 'Long Wall')

-Location: Northern China

-Length: 21,196.18 km (13,170.7 mi), all known sections were measured

-The height of the Great Wall is 5–8 meters (16–26 feet), where intact/restored. It was designed to be at least three times the height of a man. Some of the Wall was built along ridges, which make it look taller.

-History: more than 2,300 years

Why the Great Wall Was Built?

- To prevent invasion
- To protect Silk Road trade

It was an integrated military defensive system with watchtowers for surveillance, fortresses for command posts and logistics, beacon towers for communications, etc. In the Qin Dynasty, the First Emperor of Qin inked the northern walls to prevent invasion from northern nations. In the Han Dynasty, the emperors extended the Great Wall far into today's western China to protect Silk Road trade.

SCIENCE

SEISMOMETER

Zhang Heng

132 A.D

It’s function is to determine the direction of an earthquake occurring inLongxi a thousand kilometers away. It was the first time mankind has detected an earthquake.

ASTRONOMY

ECLIPSES

Chinese astronomers recorded 1,600 observations of solar and lunar eclipses from 750 BC. The ancient Chinese astronomer Shi Shen (fl. fourth century BC) was aware of the relation of the moon in a solar eclipse, as he provided instructions in his writing to predict them by using the relative positions of the moon and the sun.

TIMEKEEPING

CALENDAR

The traditional Chinese calendar is a lunisolar calendar which reckons years, months, and days according to astronomical phenomena on May 12, 2017. China now officially uses the Gregorian calendar.

LITERATURE AND PHILOSOPHY

FOUR BOOKS AND FIVE CLASSICS

Four books are Chinese classic texts illustrating the core value and belief systems in Confucianism.

- Great Learning
- Doctrine of the Mean
- Analects
- Mencius

The Five classics include
- Book of Poetry
- Book of documents
- Books of Rites
- Book of changes
- Spring and Autumn annals

DEVELOPMENT OF SCIENCE IN MEDIEVAL MIDDLE EAST

Middle East and the Golden Age of Islam,The rise of Islamic Science had its highest point between the 8th and 13th century, a period known as the Islamic Golden age.

Arab Scientists made STAGGERING breakthroughs which broadened mankind's comprehension of the natural world.

Arabic scientific inquiry flourished into major discoveries in the fields of mathematics, physics, astronomy, chemistry, medicine and optics.

The geographical location of the Middle East as a crossroad of travelers was perfect for information exchange.

Indians and Byzantinian scientist often collaborated with Arab scientist to perform studies.

House of Wisdom was built in Baghdad during the Golden Age of Islam, Baghdad was considered the center of learning in its glorious days.

Birth of the Modern Scientific Method

Muslim scientist placed greater value on science experiments rather than plain-thought experiments.

Arab scientists of the 12th century conducted experiments in their pursuit of scientific inquiry.

Using intuition, they made and tested hypotheses, and sought proof to verify their theories.

These processes would eventually become known as the modern scientific method.

Experiment is what differentiated Arabic science from Greek science (which used theory and speculation).

Today the modern scientific method consists of gathering evidence by way of reasoning, formulating hypotheses and conducting experiments.

Arabic Scientist and Their Contributions to Science and Technology

Arabic scholars who made advances in science and technology while Europe was in a cultural decline during the Dark Ages (5th – 15th century).

Ibn al-Haytham
"Father of Modern Optics"

Ibn al-Haytham was a polymath and engineer.

He discovered the laws of refraction.

Muhammad ibn Musa al-Khwarizmi

"The Father of Algebra"

- He produced a comprehensive guide to the numbering system, developed from the Brahmi system in India, using only 10 digits (0-9, the so-called 'Arabic numerals').
- Also used the word algebra ('al-jabr') to describe the mathematical operations he introduced, such as balancing equations, which helped in several day-to-day problems.
- Hisab al-Jabrwa-al Muqabala

Jabir ibn Hayyan "Father of Arab Chemistry"

- An alchemist who, in his quest to make gold from other metals,

discovered strong acids such as sulfuric, hydrochloric and nitric acids.
- He discovered the only substance that can dissolve gold – aqua regia (royal water) - a volatile mix of hydrochloric and nitric acid.

Al-Zharawi

- Considered one of the fathers of modern surgery.
- Known for the invention of many surgical instruments, including forceps to assist in vaginal births.
- He wrote Al-Tasrif: An Encyclopedia of Medicine which became the standard reference for Islamic and European medicine for over 50 years.

Ibn Sina or Avicenna

"Father of Early Modern Medicine"

- Made important contributions to the disciplines of physics, optics, philosophy and medicine.
- His works The Books of Healing and The Canon of Medicine, a medical encyclopedia became standard medical textbooks at many medieval universities.

Ibn Sina or Avicenna

The Book of Healing- also called The Cure is intended to "cure or heal" ignorance of the soul.

Four parts of the Book of Healing:
1. Logic
2. Natural science
3. Mathematics
4. Metaphysics

The Canon of Medicine- an encyclopedia of medicine in five books compiled by Ibn Sina.

The Canon of Medicine is divided into five books:
1. General Medicine Principles
2. Materia Medica
3. Special Pathology
4. Systemic Diseases
5. Pharmacopeia

Nasir al-Din al-Tusi

- Authorized the Treasury of Astronomy, an accurate table of planetary movements that reformed the existing planetary model of Roman astronomer Ptolemy by describing a uniform circular motion of all planets in their orbits.
- Copernicus later drew heavily on the work of al-Din al-Tusi and his students, but without acknowledgement.

Al-Razi

- Authored an influential multi-volume encyclopedia of all known medical knowledge in the 9th century.
- His study on smallpox and measles were groundbreaking.
- Razi had a great influence on Western medicine, and he is considered one of the greatest scientists of the Middle East.

Ibn al-Nafis "**Father of Circulatory Physiology"**

- He identified pulmonary transit – that is, that blood enters the heart at the right atrium, exits via the right ventricle, arrives at the lungs where it is reoxygenated, then passes back through the heart's left atrium and is redirected back to the body.
- Right atrium-right ventricle-pulmonary artery-lungs-pulmonary veins-left atrium
- Left atrium-left ventricle-aorta-body-vena cava-right atrium
- Prior to this, it was thought that blood simply seeped through holes

between the heart's chambers and did not pass through the lungs.

Other Important Technological Inventions of Arab Scientist and Engineers

- Brass astrolabes were developed in Iran (Persia) for navigation and for finding the quibla (the direction of Mecca).
- Mercuric Chloride – used to disinfect wounds.
- Mental Institute – hospital that provided care for the insane.
- Bridge Mill - A mill that uses hydropower.
- Trebuchet - a type of catapult, a common and powerful type of siege engine.
- Coffee - It was in Yemen that coffee beans were first roasted and brewed as they are today.
- Hookah or waterpipe – used for smoking tobacco.
- Marching band – First seen marching with the Ottoman Army.
- Among some of the technological inventions attributed to Arab scientists and engineers are the parachute, hang-glider, eye glasses, artificial wings, soft drinks, fine glass, modern soap, shampoo, kerosene, mechanical clocks.

Decline of Islamic Golden Age

- In 1206, The Mongolians led by Genghis Khan established a powerful dynasty in Asia.
- In 1258, Mongols seized Baghdad and destroyed the House of Wisdom. Legend holds that so many books were thrown into the. Tigris River that ran black with ink.
- The destruction of the House of Wisdom was seen by some as the as the end of the Islamic Golden Age.

- The Caliphate was demolished.
- Due to the conquest of the Mongolians and the crusaders, libraries, observatories, and other learning institutions were destroyed.
- Genghis Khan went on to build the largest empire the world has ever seen; he killed 10% of the global population in his quest to conquer the world.
- Had history played out differently, Islam might look very different indeed today